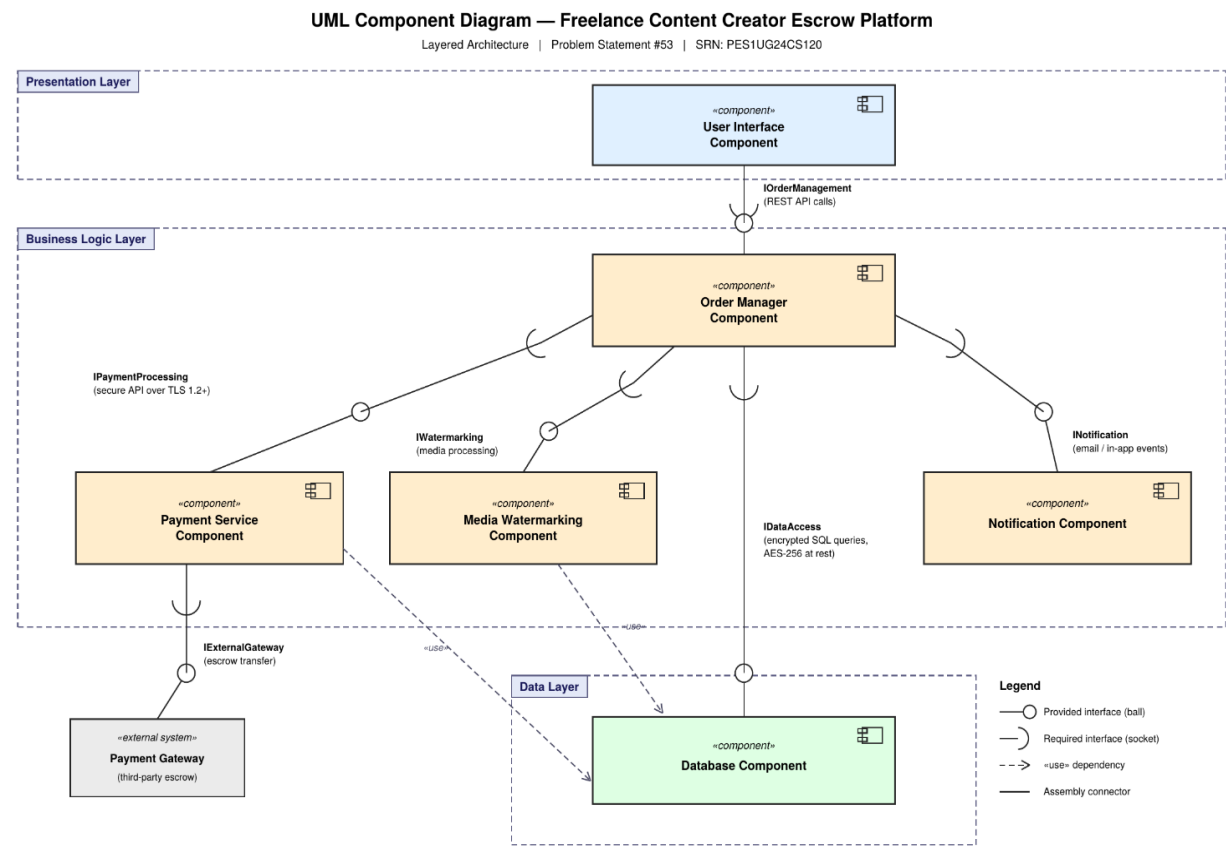


Lab 3: Component Modelling & Architectural Pattern Selection

Problem Statement #53 – Freelance Content Creator Escrow Platform

Name: Catherine
SRN: PES1UG24CS120
Sec: 5B

UML Component Diagram - Freelance Content Creator Escrow Platform



Architectural Justification

Architecture Selection

We chose Layered Architecture for the Freelance Content Creator Escrow Platform System.

Architectural Choice

The system is organized into three horizontal layers — a Presentation Layer (User Interface Component), a Business Logic Layer (Order Manager, Payment Service, Media Watermarking, and Notification Components), and a Data Layer (Database Component) — with a third-party Payment Gateway accessed only through the Payment Service Component.

Two Scenario-Related Reasons

1. Centralized, auditable business rules: The platform's core function is to release real money only after an explicit sponsor sign-off (FR-001, FR-003). A Layered Architecture forces every request — submit draft, approve, request revision, release payment — to pass through the single Order Manager Component in the Business Logic Layer, so the “no payment without approval” rule can be enforced and audited in one place rather than being duplicated or bypassed across multiple independent services.
2. Manageable complexity for a bounded workflow: The escrow workflow is a well-defined, moderately complex sequence (define milestone → submit → watermark → review → approve/revise → pay → notify), not a large system needing independent scaling of dozens of services. A Microservices approach would add operational overhead and cross-service data-consistency risk around financial transactions, while a plain Client-Server model would lack separation between UI, business rules, and storage. Layered Architecture gives the clear separation of concerns the scenario needs without the extra complexity Microservices would introduce.

Security Advantage

Because the Database Component is only reachable through the Business Logic Layer (IDataAccess) and never directly from the Presentation Layer, all financial and milestone data access is funneled through one controlled point where encryption (AES-256 at rest, TLS 1.2+ in transit — NFR-002) and authorization checks can be consistently enforced, reducing the attack surface compared to letting the UI query the database directly.

Performance Benefit

Isolating the Media Watermarking Component within its own business-layer module (IWatermarking interface) lets the automated watermarking step (which must complete in under 5 seconds per NFR-001) be optimized, cached, or scaled independently of the Order Manager's request-handling logic, without requiring changes to the UI or Payment layers.